

Indigenous Data Governance.

Improving Community Governance and Archival Practices



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Introduction

Community governance refers to the way a community makes collective decisions through community assemblies on issues such as election of local authorities, use of natural resources, allocation of financial capital, and conflict resolution, among others. Some fundamental aspects of this governance are participation, transparency, and preservation of information. The latter includes classifying, organizing, digitizing, and preserving the documentary heritage that safeguards the community's historical memory, legacy, and identity.

Background and Significance

In Mexico, there are 31 state and 2,469 municipal archives¹. However, there are no archive records at the local or community level, even though Indigenous peoples and communities constitute the foundation of the Mexican nation.

In accordance with the traditional organization of Mexico's Indigenous communities, community documents are typically kept at the community's administration headquarters, property services commissioner's office, or the local temple.



In most Indigenous communities, the documents that safeguard historical memory are disorganized and in a state of deterioration due to inadequate archival management practices.

It is essential to establish and implement data governance techniques tailored to Indigenous peoples in Mexico, that consider their own realities and interests, since these peoples have the right to manage their own data according to their worldview and priorities².

Aims

To implement a community-centered digital archive model that enables the recovery, organization, and preservation of the historical documentary heritage of Indigenous peoples through the creation of an ontology focused on the community needs, and based on governance principles³ to ensure proper preservation, access, consultation, and replicability of the model in other Indigenous communities.

Methods

Review phase

- We performed a literature review of applicable ontologies to capture essential data from indigenous community documents⁴⁻⁸.
- We reviewed 10% of the documents safeguarded at the headquarters of the community administration at San Jerónimo Purenchécuaro, Michoacán, México, between 1911 and 2025.
- We identified documents representative of the different community governance activities, as well as the essential data in each of them.

Definition phase

- We selected the InterPARES^{7,8} ontology. InterPARES is used in archival and information management, and it aligns with the requirements identified in the historical heritage of indigenous peoples' records.
- We identified relevant features from the InterPARES ontology and used them to define the archival model for this project.
- We added a metadata section to the model.⁹
- Through iterations, we refined the ontology for communal archives.
- The final ontology consists of three dimensions: Metadata, Intellectual/Conceptual Components, and Digital Components.

Model validation phase

- We validated the ontology model against 14 documents representative of the different types of writings between 1911 and 2025, to see if it captured the relevant information from the communal archives.

Results

- The resulting ontology meets the needs, worldview, and priorities of indigenous communities. It aligns with governance principles of respect for privacy, ethical aspects, and minimization of bias and discrimination.
- The model is applicable to different indigenous communities, regardless of their ethnic affiliation.
- Figure 1 shows the components that articulate the three dimensions that make up the ontology for communal archives: i) Metadata allows for sustainable content management; ii) Intellectual/conceptual components capture the set of formal attributes of the physical document; and iii) Digital components capture the attributes of the electronic or digitized archival document.

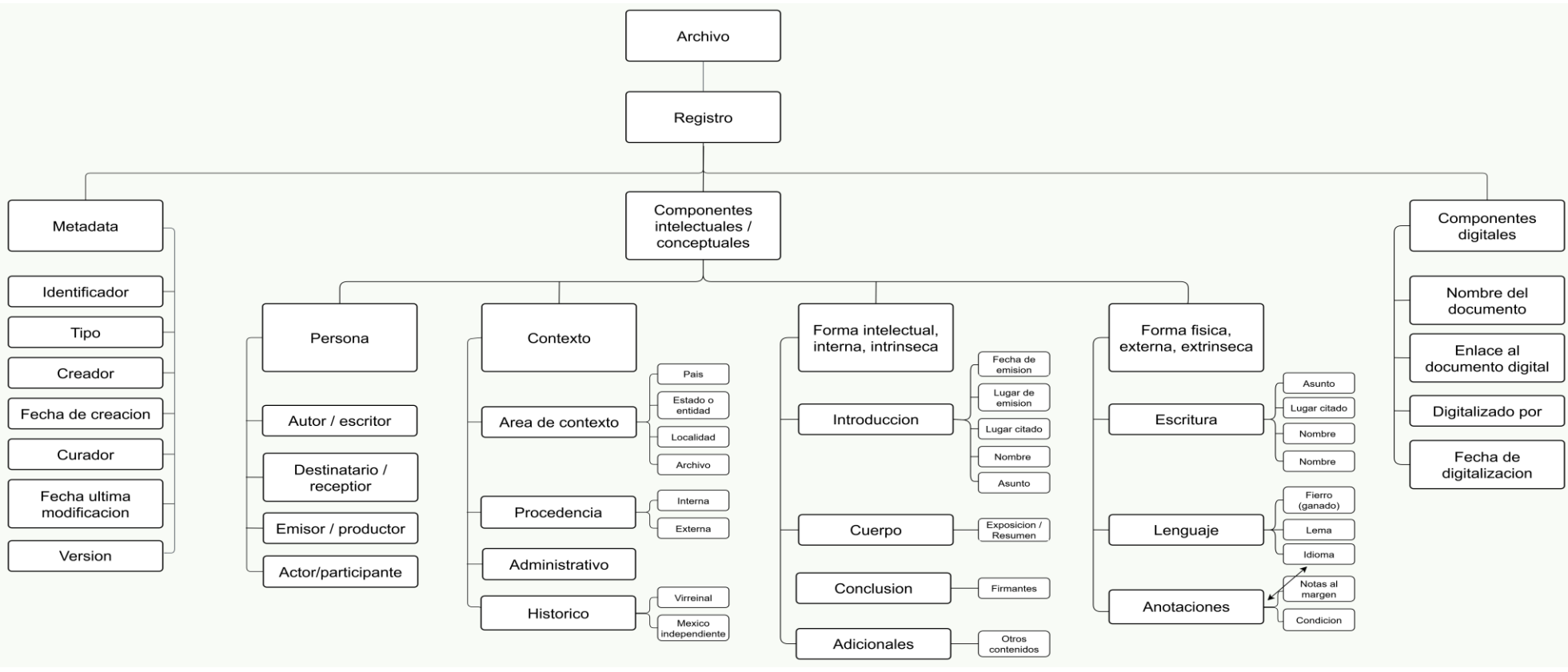


Figure 1: Ontology for Communal Archives

Discussion

- The use of ontologies, governance and data infrastructure, in combination with Indigenous communities' knowledge and laws, facilitates the development of data governance mechanisms designed by and for Indigenous peoples.
- This ensures that technologies, and the policies that regulate their use, uphold the rights of Indigenous peoples, integrate their values, support their self-determined future, and contribute to the preservation of their historical memory.
- An ontology offers significant advantages for management, retrieval, and preservation of community's archives. It supports portability to various platforms, and reusability of content without the need to recreate a model each time.

Future work

- Continue the communal archive digitization process to preserve the historical memory of the community, with information duly ordered and classified.
- Implement the ontology model to formalize the digitalization process, to record the corresponding knowledge and information.
- Replicate the low-cost digital archive model in other indigenous communities to support their cultural heritage.

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